

Name: Sk Arshad Basha

Reg No :192312336

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

Introduction

A retail company can develop a product recommendation system using K-Nearest Neighbour (KNN). KNN recommends products by finding customers whose purchasing behaviour is most similar to the target customer. The products purchased by these similar customers can then be recommended.

KNN is a supervised, instance-based machine learning algorithm. It does not build a complicated mathematical model during training. Instead, it stores historical customer data and compares a new customer with existing customers.

The recommendation system works as follows:

Customer Purchase History



Feature Extraction



Feature Normalization



Calculate Customer Similarity



Find K Nearest Customers



Collect Their Purchased Products



Remove Already Purchased Products



Rank Products



Generate Recommendations

Adapting to New Customer Data

The system can continuously update itself.

When a customer makes a new purchase:

1. The transaction is stored.
2. Customer purchase features are updated.
3. Similarity calculations use the updated information.
4. Future recommendations change according to the latest behaviour.

Thus, the system becomes more personalized over time.

Improving Accuracy

Accuracy can be improved by:

- Normalizing numerical features.
- Selecting relevant purchase features.
- Testing different values of K.
- Giving greater importance to recent purchases.
- Removing duplicate or incorrect transactions.
- Using cross-validation.
- Combining KNN with content-based recommendation.

Advantages

- Simple to understand.
- Easy to implement.
- Adapts to new customers.
- Does not require complicated model training.
- Can provide personalized recommendations.

Limitations

- Searching a very large customer database can be computationally expensive.
- Irrelevant features can reduce recommendation quality.
- The choice of K affects the results.
- New customers with no purchase history create a cold-start problem.

Conclusion

KNN is suitable for a retail recommendation system because it identifies customers with similar purchasing behaviour and recommends products based on their previous purchases. Continuous addition of new transaction data allows recommendations to adapt to changing customer preferences.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

Introduction

A manufacturing company needs to predict machine output from sensor readings that change over time. Locally Weighted Regression (LWR) is suitable because it gives greater importance to historical observations that are close to the current machine condition. Locally Weighted Regression is a regression technique in which each training observation receives a weight according to its distance from the query point. Unlike ordinary linear regression, which creates one global model, LWR creates a model that focuses on the neighbourhood of the current query.

Possible sensor inputs are:

- Temperature
- Pressure
- Vibration
- Speed
- Voltage
- Current

Proposed System

Sensor Data



Data Cleaning



Feature Selection



New Sensor Reading



Calculate Distance



Calculate Gaussian Weights



Build Local Regression Model



Predict Machine Output

Why Local Models Improve Accuracy

Machine behaviour is not always the same.

For example:

- At low temperature, machine output may increase slowly.
- At normal temperature, output may be stable.
- At very high temperature, output may decrease.

A single global regression equation may not represent all these conditions.

LWR concentrates on historical observations similar to the current operating condition. Therefore, it can provide a more relevant local prediction.

Handling Noisy Data

Manufacturing sensors may contain noise.

The system can use:

- Moving-average filtering.

- Median filtering.
- Outlier detection.
- Sensor calibration.
- Data smoothing.

Advantages

- Captures local behaviour.
- Useful for changing machine conditions.
- Flexible compared with global regression.
- Can adapt to nonlinear trends.

Limitations

- Prediction can be computationally expensive for large datasets.
- Performance depends on bandwidth selection.
- Sensitive to poor-quality data.
- Requires sufficient historical observations around the query.

Conclusion

LWR is an effective solution for manufacturing because it focuses on historical sensor readings similar to the current machine condition. It can improve prediction accuracy when machine behaviour changes over time.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

Introduction

A healthcare company can use a Radial Basis Function (RBF) Network to classify patients into different risk categories such as Low Risk, Medium Risk, and High Risk. An RBF network is a type of neural network consisting mainly of:

The hidden layer uses radial basis functions, commonly Gaussian functions.

Possible patient features include:

- Age

- Blood pressure
- Blood glucose
- Cholesterol
- BMI
- Heart rate

Proposed System

Historical Medical Data



Data Cleaning



Feature Selection



Normalization



Select RBF Centres



Gaussian Activation



Train Output Layer



New Patient Data



Risk Classification

Example

Suppose the system receives:

Age = 60

Blood Pressure = High

Glucose = High

BMI = 31

The RBF network compares this patient's feature pattern with learned centres.

If the output scores are:

Risk	Score
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Low	0.10
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Medium	0.25
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High	0.65
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The system predicts:

High Risk

In a real healthcare system, such predictions should support—not replace—clinical judgment.

Feature Selection

Not every feature is equally useful.

Important features can be selected using:

- Correlation analysis.
- Feature importance.
- Statistical tests.
- Domain/clinical knowledge.
- Recursive feature elimination.

Irrelevant features can increase computational cost and reduce accuracy.

Improving Accuracy

Accuracy can be improved through:

- Feature normalization.
- Better RBF centre selection.
- Appropriate Gaussian width.
- Cross-validation.

- Hyperparameter tuning.
- Handling class imbalance.
- Removing noisy records.

Evaluation should include:

Advantages

- Handles nonlinear relationships.
- Can provide fast prediction after training.
- Suitable for classification.
- Can model complex medical patterns.

Limitations

- Requires appropriate centre selection.
- Large datasets may require efficient implementation.
- Poor-quality medical data can reduce performance.
- Predictions require careful validation before clinical use.

Conclusion

An RBF network is suitable for patient risk classification because it can learn complex nonlinear relationships between medical features and risk categories.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.

Introduction

An e-commerce company can use Case-Based Learning (CBL) to identify products similar to a new product and reuse historical pricing information for dynamic pricing. Case-Based Learning solves a new problem by retrieving and adapting solutions from previously stored cases.

Each product is stored as a case containing information such as:

- Product category
- Brand
- Price

- Size
- Colour
- Material
- Specifications
- Rating
- Previous sales

When a new product arrives, it is compared with existing cases.

Proposed System

New Product



Extract Product Features



Compare With Historical Cases



Calculate Similarity



Retrieve Similar Products



Retrieve Previous Prices/Sales



Determine Suitable Price



Store New Case

Scaling with Increasing Product Data

When the number of products increases, searching every product becomes expensive.

The system can use:

- Product-category filtering.

- Feature indexing.
- Database indexing.
- Approximate nearest-neighbour search.
- Duplicate removal.
- Product segmentation.

Advantages

- Reuses previous experience.
- Easy to update.
- No complete retraining is required.
- Suitable for dynamic environments.
- Provides interpretable similar-product results.

Limitations

- Requires good feature representation.
- Poor similarity definitions produce poor results.
- Large case bases can increase search time.
- Old cases may become less relevant.

Conclusion

Case-Based Learning is suitable for e-commerce product similarity because it retrieves similar historical products and reuses their pricing and sales information to support dynamic pricing decisions.

5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

Introduction

A logistics company can use an RBF Network to predict delivery time using location, distance, traffic, and other environmental factors. Delivery time is influenced by nonlinear relationships, making RBF Networks suitable for this application.

Input Features

The system can use:

- Pickup location
- Delivery location
- Distance
- Traffic level
- Time of day
- Day of week
- Weather
- Vehicle type

Proposed System

Historical Delivery Data



Data Cleaning



Feature Extraction



Normalization



RBF Centres



Gaussian Activation



Train Network



Current Location + Traffic



Predicted Delivery Time

Example

Suppose a new delivery has:

Distance = 15 km

Traffic = Heavy

Time = 6:00 PM

The RBF network finds historical deliveries with similar conditions and uses their learned relationships to estimate delivery time.

For example:

Predicted delivery time = 58 minutes

The prediction can then be updated if new traffic information becomes available.

Improving Reliability

Reliability can be improved by:

- Adding recent delivery records.
- Updating traffic information.
- Removing abnormal historical records.
- Selecting suitable RBF centres.
- Tuning Gaussian widths.
- Monitoring prediction errors.
- Retraining periodically.

Advantages

- Captures nonlinear relationships.
- Handles multiple input variables.
- Learns complex traffic patterns.
- Can adapt to changing conditions.
- Provides fast predictions after training.

Limitations

- Requires sufficient historical data.
- Centre and width selection affect performance.
- Unexpected events can cause prediction errors.
- Regular updating is required.

Conclusion

An RBF network is an effective solution for logistics because it can model nonlinear relationships between distance, location, traffic, time, and delivery duration. Continuous updates with recent delivery and traffic data improve prediction reliability.